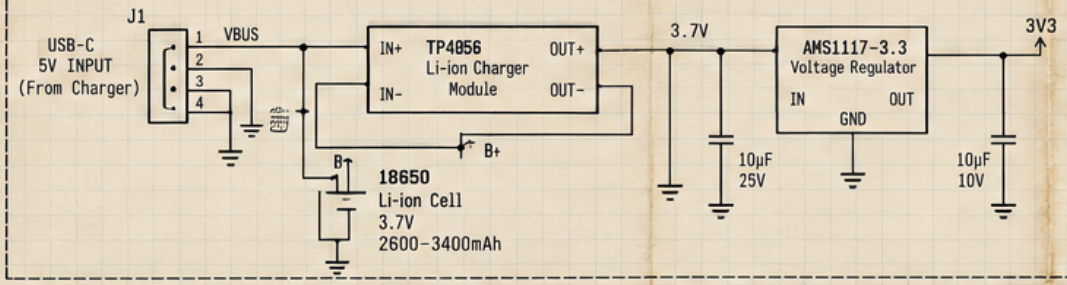
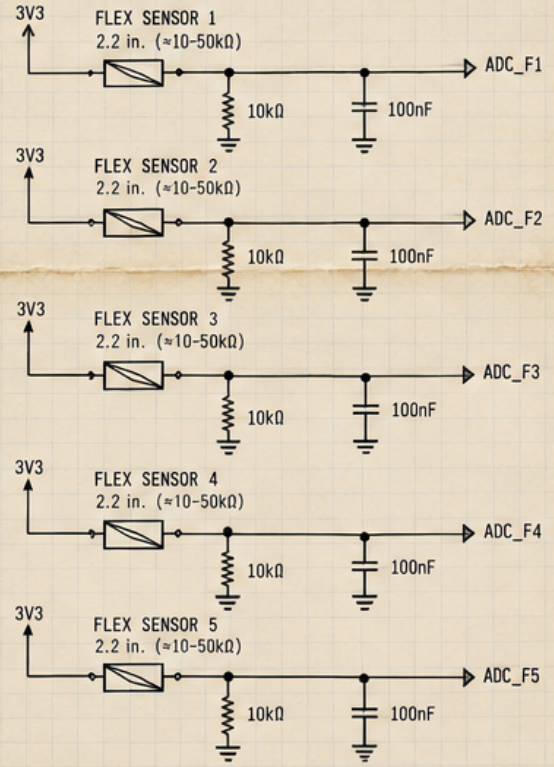
1. POWER SUPPLY

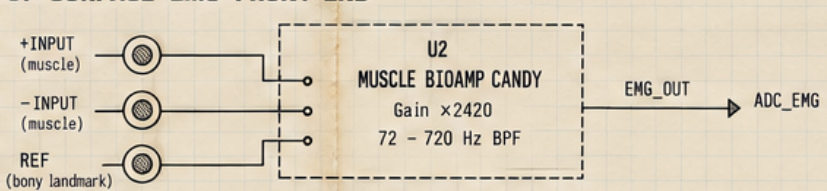


2. FLEX SENSOR VOLTAGE DIVIDERS — 5 CHANNELS



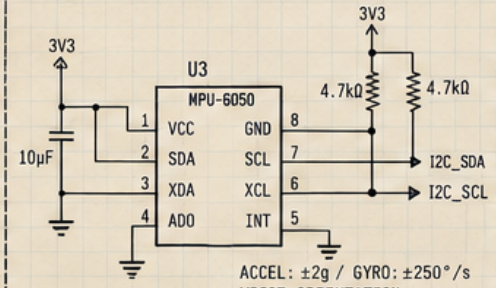
NOTES:
1. Flex sensors are variable resistors (≈10kΩ–50kΩ).
2. 10kΩ pull-down forms voltage divider with 3V3.
3. 100nF capacitor for noise filtering.
4. ADC range: 0 – 3.3V (ESP32-S3 ADC).

3. SURFACE EMG FRONT-END



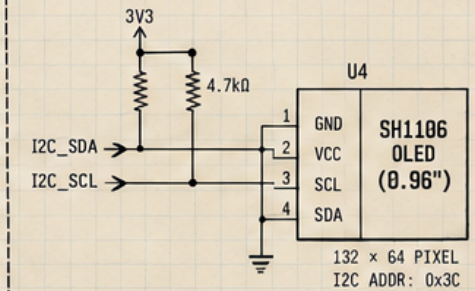
3-ELECTRODE SURFACE EMG – FOREARM FLEXOR

4. 6-AXIS IMU (MPU-6050)



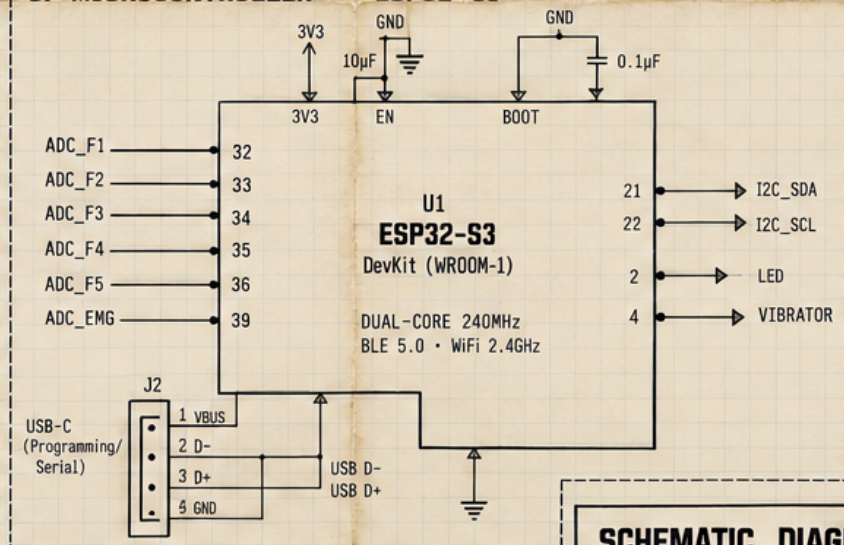
ACCEL: ±2g / GYRO: ±250°/s
WRIST ORIENTATION

5. SH1106 OLED DISPLAY



132 × 64 PIXEL
I2C ADDR: 0x3C

6. MICROCONTROLLER — ESP32-S3

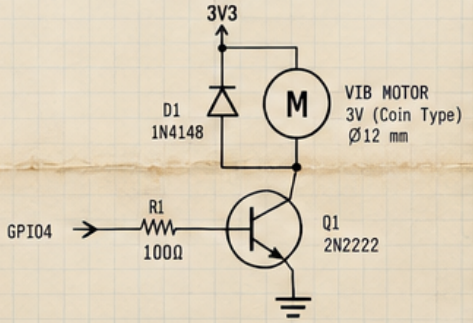


DECOUPLING (Near ESP32-S3)

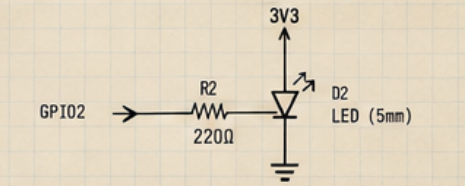


7. FEEDBACK ACTUATORS

7.1 VIBRATION MOTOR (HAPTIC FEEDBACK)



7.2 STATUS LED (VISUAL INDICATOR)



SCHEMATIC DIAGRAM — MULTIMODAL SENSING SYSTEM

PROJECT:	DWG NO:	SHEET	REV
REHABGRIP RG-2026	RG-2026-EE-001	1 OF 1	A
UNITS: —	SCALE: —	DATE: 2026	